🌐 Kubernetes Lab Challenge: Private Service with External DNS (Route 53)

🎯 Objective

Expose a Kubernetes Service **internally** within your AWS VPC using **External DNS** integrated with **Route 53**, without making it publicly available on the internet.

🧭 Challenge Description

You are working on an internal tool that must be accessed **only within your private VPC** (not over the public internet). Your task is to:
- Deploy a basic service (e.g., an NGINX or Flask app)
- Create a private internal LoadBalancer
- Automatically register its DNS in a private **Route 53 Hosted Zone** using **ExternalDNS**

💡 What You’ll Learn

- How Kubernetes integrates with AWS Route 53 using ExternalDNS
- How to expose services **privately** using internal LoadBalancers
- Controlling DNS and traffic visibility within VPC

🔧 Tasks

🟢 Step 1: Create a Private Hosted Zone in Route 53
- Use AWS Console or CLI to create a **Private Hosted Zone** (e.g., `internal.example.com`)
- Associate it with your VPC

📘 Reference:
- https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/hosted-zones-private.html

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🟢 Step 2: Deploy ExternalDNS with IAM Permissions
- Deploy ExternalDNS as a Helm chart or YAML manifest
- Ensure it is configured for **private hosted zone** access
- IAM role (or IRSA) must allow:
  - `route53:ChangeResourceRecordSets`
  - `route53:ListHostedZones`

📘 Reference:
- https://github.com/kubernetes-sigs/external-dns/blob/master/docs/tutorials/aws.md

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🟢 Step 3: Deploy an Internal Service
- Create a Kubernetes **Service of type LoadBalancer**
- Add the following annotations:
```yaml
  service.beta.kubernetes.io/aws-load-balancer-internal: "true"
  external-dns.alpha.kubernetes.io/hostname: "nginx.internal.example.com"
```

📘 Reference:
- https://docs.aws.amazon.com/eks/latest/userguide/load-balancing.html
- https://github.com/kubernetes-sigs/external-dns/blob/master/docs/tutorials/aws.md

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🟢 Step 4: Test from Within VPC

- Launch a test pod or EC2 instance within the same VPC
- Verify DNS resolution using:
```bash
dig nginx.internal.example.com
curl http://nginx.internal.example.com
```

✅ You should **NOT** be able to access it from outside the VPC.

✅ Completion Criteria

- `external-dns` pod is running and registered the DNS
- Route 53 private record is created
- LoadBalancer is internal (not internet-facing)
- App is accessible only from within the VPC

📦 Bonus Tasks (Optional)

- Deploy `CoreDNS` + `NodeLocal DNSCache` to speed up DNS
- Apply network policy to limit access to specific namespaces
- Configure IRSA for ExternalDNS instead of using access keys

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🚀 Submission Instructions

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Be ready to:
- Demonstrate DNS registration
- Show AWS Load Balancer type
- Confirm internal-only access
```